Quantitative Research – Mock Exam 1

Solutions, results overview and grading rubric

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Structure. Section 1 lists all 30 final results on one page (quick grading pass). Then, for every subtask, follow the worked solution, the highlighted final result and a *grading rubric* for the 2 points.

1 Results overview – all final results at a glance

Grading note. Each subtask asks for exactly *one* result: a correct result with comprehensible working earns 2 points (split for partial answers: see the rubric of the subtask). A result without working, or correct working with a mere slip (transposed digits, rounding), counts 1 point. All subtasks are independent – there are no follow-on errors.

Part	Asked for	Result	P.
Task 1 – Statistical inference: z-test and confidence interval			20 points
a)	Hypothesis pair	$H_0\colon \mu = 80$ vs. $H_1\colon \mu \neq 80$	2
b)	Standard error	$\sigma/\sqrt{n} = 12/6 = 2$ EUR	2
c)	Test statistic Z_n	$Z_n = 4.30/2 = 2.15$	2
d)	Python: critical value with `stats.norm.ppf`	$\mathtt{q} = 1 - \alpha/2 = 0.975$ (returns $z_{0.975} = 1.96$)	2
e)	Test decision	$\lvert Z_n\rvert = 2.15 > 1.96 \Rightarrow$ reject H_0	2
f)	p-value	$p = 2\,(1 - 0.9842) = 0.0316$	2
g)	95 % confidence interval	$84.30 \pm 3.92 = [80.38;\ 88.22]$ EUR	2
h)	Interpretation of the CI	with 95 % confidence, the mean μ of *all* orders lies in [80.38; 88.22] EUR	2
i)	Type I error in context	rejecting H_0 ("mean changed") although $\mu = 80$ EUR still holds	2
j)	Python command (t-test)	`stats.ttest_1samp(orders, popmean=80)`	2
Task 2 – Simple and multiple linear regression			20 points
a)	Slope $\hat{\beta}_1$	$\hat{\beta}_1 = 120/40 = 3$	2
b)	Intercept $\hat{\beta}_0$	$\hat{\beta}_0 = 60 - 3 \cdot 24 = -12$	2
c)	Interpretation of the slope	1 °C warmer $\Rightarrow$ on average about 3 more iced coffees sold	2
d)	Python: prediction with `reg.predict`	$\hat{y}(25) = -12 + 3 \cdot 25 = 63$ iced coffees	2
e)	Residual of day 5	$\hat{u}_5 = 68 - 72 = -4$	2
f)	Coefficient of determination R^2	$R^2 = 1 - 40/400 = 0.9$	2
g)	Correlation coefficient r_{xy}	$r_{xy} = +\sqrt{0.9} = 0.9487$	2
h)	Python command (OLS)	`smf.ols("sales ~ temp", data=cafe).fit()`	2
i)	Ceteris-paribus interpretation	holding rain fixed: $+1$ °C $\Rightarrow$ on average 3.01 more iced coffees	2
j)	Significance of `rain`	$p = 0.253 > 0.05 \Rightarrow$ not significant (keep $H_0\colon \beta_2 = 0$)	2
Task 3 – Classification: logistic regression and evaluation			20 points
a)	Odds of churn	odds $= 0.2/0.8 = 0.25$	2
b)	Churn probability at $x = 2$	$\hat{\eta} = -2,\ \hat{p} = 0.1353/1.1353 = 0.1192$	2
c)	Interpretation of the odds ratio	each extra call multiplies the *odds* of churn by 1.65 (+65 %)	2
d)	Classification at $x = 7$	$\hat{\eta} = 0.5 > 0,\ \hat{p} = 0.6225 > 0.5 \Rightarrow \hat{y} = 1$ (churn)	2
e)	Python command (logit)	`smf.logit("churn ~ calls", data=df).fit()`	2
f)	Accuracy	$(30 + 140)/200 = 0.85$	2
g)	Python: row shares with `pd.crosstab`	row share $30/(30 + 10) = 0.75$ (the sensitivity)	2
h)	Specificity	$140/(140 + 20) = 0.875$	2
i)	Precision	$30/(30 + 20) = 0.6$	2
j)	Lower threshold $t = 0.3$	sensitivity **increases** (more churners caught)	2
		Total	**60**

2 Task 1 – Statistical inference: z-test and confidence interval (20 points)

Scenario. Two-sided z-test (σ known) of the mean order value of an online shop: $\mu_0 = 80$ EUR, $\sigma = 12$ EUR, $n = 36$, $\bar{x} = 84.30$ EUR, $\alpha = 0.05$. Asked for are the test in its steps, the p-value, the 95 % confidence interval, two interpretations and two Python questions (the quantile function `stats.norm.ppf` and the command for the t-test).

↪ *Lecture reference:* Part 3, sections *Confidence Intervals* (known variance), *Hypothesis Tests*, *The p-Value* and *Types of Error*; Python: slide "Hypothesis test with Python – `ttest_1samp`" and Summary 6 (`stats.norm.ppf`).

The concepts behind this task

- **Standard error** (σ known): $\mathrm{SE} = \sigma/\sqrt{n}$ – how strongly $\bar{X}$ fluctuates around μ.
- **Test statistic:** $Z_n = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$ under H_0; two-sided: reject H_0 if $|Z_n| > z_{1-\alpha/2}$.
- **p-value** (two-sided): $p = 2\,\Phi(-|Z_n|)$; reject H_0 if $p < \alpha$.
- **Confidence interval:** $\mathrm{CI}_{1-\alpha} = \left[\bar{X} \pm z_{1-\alpha/2}\,\sigma/\sqrt{n}\right]$; μ_0 outside the CI $\Leftrightarrow$ reject H_0 (duality).
- **Type I error:** reject H_0 although H_0 is true (false alarm); its probability is α.

a) Hypothesis pair

The management wants to find out whether the mean order value after the redesign differs *from the planning figure of 80 EUR (in either direction). State the hypothesis pair.*

The statement the management wants to find evidence for – the mean order value *differs* from the plan – goes into H_1. Because a deviation in either direction counts, the test is two-sided:

$$H_0 : \mu = 80 \qquad \text{vs.} \qquad H_1 : \mu \neq 80.$$

Here μ is the unknown mean order value of *all* orders after the redesign. Hypotheses are always statements about the population parameter, never about $\bar{x}$.

Result: $H_0 : \mu = 80$ vs. $H_1 : \mu \neq 80$ (two-sided)

Grading (2 P.): 1 P. $H_1 : \mu \neq 80$ (two-sided alternative), 1 P. matching $H_0 : \mu = 80$.

Typical mistake: hypotheses written with $\bar{x}$ instead of μ, or a one-sided H_1 although both directions matter.

b) Standard error

Compute the standard error $\sigma/\sqrt{n}$ of the sample mean for $\sigma = 12$ EUR and $n = 36$.

$$\mathrm{SE} = \frac{\sigma}{\sqrt{n}} = \frac{12}{\sqrt{36}} = \frac{12}{6} = 2 \text{ EUR}.$$

The mean of 36 orders fluctuates much less than a single order value: its standard deviation is only one sixth of σ.

Result: $\dfrac{\sigma}{\sqrt{n}} = \dfrac{12}{\sqrt{36}} = \dfrac{12}{6} = 2$ EUR

Grading (2 P.): 1 P. formula with $\sqrt{36} = 6$, 1 P. result 2.

Typical mistake: dividing by n (0.33) or by $\sqrt{n-1}$ instead of $\sqrt{n}$.

c) Test statistic Z_n

Compute the test statistic Z_n for $\bar{x} = 84.30$ EUR, $\mu_0 = 80$ EUR, $\sigma = 12$ EUR and $n = 36$.

$$Z_n = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{84.30 - 80}{12/\sqrt{36}} = \frac{4.30}{2} = 2.15.$$

The sample mean lies 2.15 standard errors *above* the hypothesised value $\mu_0 = 80$ EUR.

Result: $Z_n = \dfrac{84.30 - 80}{12/\sqrt{36}} = \dfrac{4.30}{2} = 2.15$

Grading (2 P.): 1 P. formula with correct substitution, 1 P. result 2.15.

d) Python: critical value with `stats.norm.ppf`

In Python, the critical value of the two-sided test at the significance level $\alpha = 0.05$ (σ known) is computed with the line below. Which number must be passed as `q`?

```
stats.norm.ppf(q)
```

`stats.norm.ppf(q)` is the quantile function of $N(0,1)$: it returns the value z_q with $\Phi(z_q) = q$ – the Python counterpart of searching q *inside* Table 1. Two-sided test: α is split into two tails of $\alpha/2 = 0.025$ each, so the upper critical value is the $1 - \alpha/2 = 0.975$ quantile:

```
from scipy import stats
stats.norm.ppf(0.975)      # 1.959964   ->   z_0.975 = 1.96
```

Table 1 gives the same value (row 1.9, column 0.06). H_0 is rejected if $|Z_n| > 1.96$, i.e. if $Z_n < -1.96$ or $Z_n > 1.96$.

Result: `q` $= 1 - \alpha/2 = 0.975$: `stats.norm.ppf(0.975)` returns the critical value $z_{0.975} = 1.96$

Grading (2 P.): 1 P. two-sided: α split into two tails, quantile $1 - \alpha/2$; 1 P. value 0.975 (also accepted: 0.025 with the remark that -1.96 gives the bounds ± 1.96).

Typical mistake: `q = 0.95` (one-sided quantile 1.645) or `q = 0.05` – `ppf` expects the probability p of the quantile, not α.

e) Test decision

The test statistic is $Z_n = 2.15$ and the critical value is $z_{0.975} = 1.96$. Make the test decision for $H_0: \mu = 80$ vs. $H_1: \mu \neq 80$ at $\alpha = 0.05$.

Rejection region of the two-sided test: $|Z_n| > z_{0.975}$. Here

$$|Z_n| = 2.15 \;>\; 1.96 = z_{0.975},$$

so the test statistic lies in the rejection region and $H_0: \mu = 80$ is rejected at the 5 % level. The data provide evidence that the mean order value after the redesign differs from 80 EUR – the sample points to a *higher* value ($\bar{x} = 84.30$).

Result: Since $|Z_n| = 2.15 > 1.96 = z_{0.975}$, H_0 is **rejected**: the mean order value differs significantly from 80 EUR.

Grading (2 P.): 1 P. comparison $|Z_n| = 2.15 > 1.96$, 1 P. decision "reject H_0".

f) p-value

Compute the two-sided p-value for the test statistic $Z_n = 2.15$ (use Table 1).

For $H_1: \mu \neq \mu_0$ both tails count:

$$p = 2\,\Phi(-|Z_n|) = 2\,\big(1 - \Phi(2.15)\big).$$

Table 1, row 2.1, column 0.05: $\Phi(2.15) = 0.9842$. Hence

$$p = 2\,(1 - 0.9842) = 2 \cdot 0.0158 = 0.0316.$$

Because $p = 0.0316 < 0.05 = \alpha$, the p-value leads to the same decision as the critical value: reject H_0. (Python: `2 * stats.norm.sf(2.15)` gives 0.0316.)

Result: $p = 2\,\Phi(-|Z_n|) = 2\,\big(1 - \Phi(2.15)\big) = 2\,(1 - 0.9842) = 0.0316$

Grading (2 P.): 1 P. $\Phi(2.15) = 0.9842$ from the table, 1 P. doubling and result 0.0316.

Typical mistake: forgetting the factor 2 (0.0158) or reporting $\Phi(2.15) = 0.9842$ itself.

g) 95 % confidence interval

Compute the 95 % *confidence interval for* μ *with* $\bar{x} = 84.30$ *EUR,* $\sigma/\sqrt{n} = 2$ *EUR and* $z_{0.975} = 1.96$.

$$\mathrm{CI}_{0.95} = \left[\bar{x} - z_{0.975}\frac{\sigma}{\sqrt{n}};\ \bar{x} + z_{0.975}\frac{\sigma}{\sqrt{n}}\right], \qquad e = 1.96 \cdot 2 = 3.92$$

$$\mathrm{CI}_{0.95} = [84.30 - 3.92;\ 84.30 + 3.92] = [80.38;\ 88.22]\ \text{EUR}.$$

Duality: the planning figure $\mu_0 = 80$ lies *outside* the interval – this agrees with the rejection of $H_0\colon \mu = 80$ by the two-sided test at $\alpha = 0.05$.

Result: $84.30 \pm 1.96 \cdot 2 = 84.30 \pm 3.92$, $\mathrm{CI}_{0.95} = [80.38;\ 88.22]$ EUR

Grading (2 P.): 1 P. formula $\bar{x} \pm z_{0.975}\,\sigma/\sqrt{n}$ (margin of error 3.92), 1 P. both bounds correct.

h) Interpretation of the CI

For the mean order value μ*, the* 95 % *confidence interval* [80.38; 88.22] *EUR was computed. Interpret this interval in one sentence.*

The lecture gives three correct readings: *coverage* (the procedure yields intervals that contain the fixed true μ in 95 % of all samples), *compatibility* (the interval holds the values of μ that are consistent with the data) and *duality* (it holds exactly the μ_0 a two-sided test at $\alpha = 0.05$ would not reject).

Model answer: "With 95 % confidence, the mean order value of all orders after the redesign lies between 80.38 and 88.22 EUR."

Not correct: "95 % of the orders lie in the interval" (the CI is about the *mean*, not about single orders). Strictly speaking also wrong: "μ lies in *this* interval with probability 95 %" (μ is fixed; the bounds are random) – 1 P.

Result: With 95 % confidence, the true mean order value μ of *all* orders after the redesign lies between 80.38 and 88.22 EUR – intervals constructed in this way contain the true μ in 95 % of all samples.

Grading (2 P.): 1 P. statement about the true mean μ of all orders (not about single orders or $\bar{x}$), 1 P. correct reading of the 95 % (confidence in the procedure / values compatible with the data).

Typical mistake: "95 % of all orders have a value between 80.38 and 88.22 EUR" (0 P.).

i) Type I error in context

Describe in one sentence what a Type I error would mean in this situation.

Type I error = reject H_0 although H_0 is true (false alarm):

$$P(\text{reject } H_0 \mid H_0 \text{ true}) = \alpha = 0.05.$$

In this situation: the shop concludes that the redesign has changed the mean order value, although the mean of all orders is in fact still 80 EUR. Because H_0 was rejected here, a Type I error is the

only error the test can have made. (A Type II error would be: keeping H_0 although the mean has changed.)

Result: A Type I error means that the shop concludes that the mean order value has changed (rejects H_0: $\mu = 80$) although it is in fact still 80 EUR – a false alarm, whose probability is $\alpha = 0.05$.

Grading (2 P.): 1 P. "reject H_0 / conclude a change", 1 P. "although H_0 is true ($\mu = 80$ EUR)" – in the context of the order value.

Typical mistake: describing the Type II error (keeping H_0 although the mean has changed).

j) Python command (*t*-test)

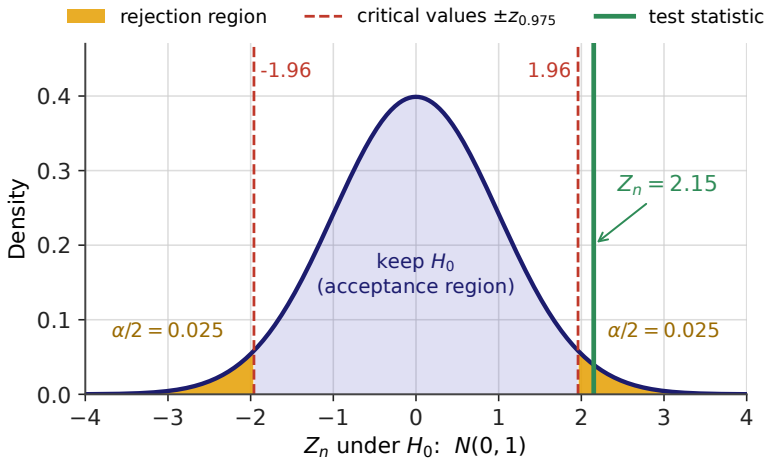
In practice σ is often unknown. Suppose the 36 order values are stored in a NumPy array `orders` *and σ is estimated from the data. Give the* `scipy.stats` *command that carries out the two-sided one-sample t-test of H_0: $\mu = 80$.*

```
from scipy import stats
res = stats.ttest_1samp(orders, popmean=80)    # two-sided by default
res.statistic, res.df, res.pvalue              # T_n, n - 1 = 35, p-value
```

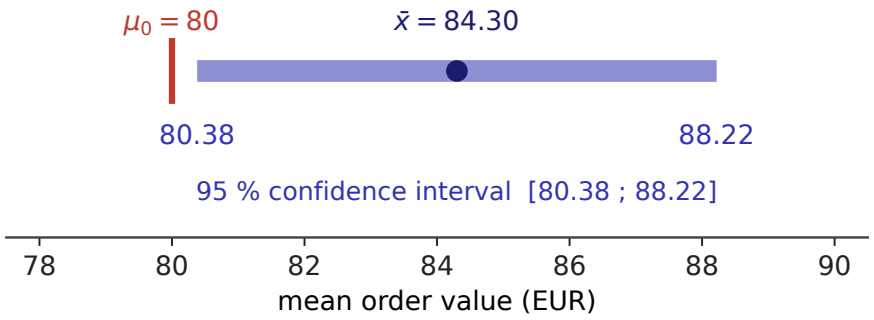
`ttest_1samp` estimates σ by s and compares T_n with the t-distribution with $n - 1 = 35$ degrees of freedom. The default `alternative="two-sided"` matches H_1: $\mu \neq 80$; one-sided tests would need `alternative="less"` or `"greater"`.

Result: `stats.ttest_1samp(orders, popmean=80)`

Grading (2 P.): 1 P. function `stats.ttest_1samp`, 1 P. arguments `orders` and `popmean=80` (also accepted: `stats.ttest_1samp(orders, 80)`, the prefix `scipy.stats.`, with or without `alternative="two-sided"`).



Standard normal distribution under H_0: rejection region $|Z_n| > 1.96$ (orange, $\alpha/2 = 0.025$ per tail) and the observed $Z_n = 2.15$.



95 % confidence interval $[80.38;\ 88.22]$: the planning figure $\mu_0 = 80$ lies outside $\Rightarrow$ reject H_0.

Typical mistakes in Task 1

- Using the one-sided quantile $z_{0.95} = 1.645$ for the two-sided test – in Python `stats.norm.ppf(0.95)` instead of `stats.norm.ppf(0.975)`.
- Forgetting the factor 2 in the two-sided p-value (0.0158 instead of 0.0316).
- Standard error with n or $n-1$ instead of $\sqrt{n}$.
- Reading the CI as the range of single order values, or as "95 % probability for *this* interval".
- Mixing up the Type I error (false alarm: reject a true H_0) and the Type II error (miss).

3 Task 2 – Simple and multiple linear regression (20 points)

Scenario. Part A: simple regression of the number of iced coffees sold (y) on the daily maximum temperature (x, in °C), $n = 5$; $\bar{x} = 24$, $\bar{y} = 60$, $S_{xx} = 40$, $S_{xy} = 120$, $S_{yy} = 400$. Part B: `statsmodels` output of the multiple regression `sales ~ temp + rain` for $n = 60$ days.

↪ *Lecture reference:* Part 4, sections *Simple Linear Regression* (least squares, interpretation), *Goodness of Fit: the Coefficient of Determination* and *Multiple Linear Regression* (ceteris paribus, t-test of a coefficient); Python: slide "Solution (2/2) – a falling price–demand line" (`reg.predict`) and Summary 5.

The concepts behind this task

- **OLS estimators:** $\hat{\beta}_1 = S_{xy}/S_{xx}$, $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$; line $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$.
- **Residual:** $\hat{u}_i = y_i - \hat{y}_i$ – positive: point above the line, negative: below.
- **Sum-of-squares decomposition:** TSS = ESS + RSS, $R^2 = \text{ESS}/\text{TSS} = 1 - \text{RSS}/\text{TSS}$.
- **Correlation** (simple regression): $R^2 = r_{xy}^2$, so $r_{xy} = \pm\sqrt{R^2}$ with the sign of $\hat{\beta}_1$.
- **Multiple regression:** $\hat{\beta}_j$ = change of y per unit of x_j *holding the other variables fixed*; $H_0\colon \beta_j = 0$ is rejected if $p < \alpha$.

a) Slope $\hat{\beta}_1$

Compute the slope $\hat{\beta}_1$ from $S_{xy} = 120$ and $S_{xx} = 40$.

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{120}{40} = 3.$$

The slope has the sign of S_{xy}: the cloud of points rises.

Result: $\hat{\beta}_1 = \dfrac{S_{xy}}{S_{xx}} = \dfrac{120}{40} = 3$ (iced coffees per °C)

Grading (2 P.): 1 P. formula S_{xy}/S_{xx}, 1 P. result 3.

Typical mistake: inverted ratio $S_{xx}/S_{xy} = 0.33$.

b) Intercept $\hat{\beta}_0$

Compute the intercept $\hat{\beta}_0$ from $\hat{\beta}_1 = 3$, $\bar{x} = 24$ and $\bar{y} = 60$.

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\,\bar{x} = 60 - 3 \cdot 24 = 60 - 72 = -12.$$

Estimated line: $\hat{y} = -12 + 3x$; it passes through the centroid $(\bar{x}, \bar{y}) = (24;\ 60)$. The intercept has no sensible meaning of its own: $x = 0$ °C lies far outside the observed range 20–28 °C (extrapolation).

Result: $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x} = 60 - 3 \cdot 24 = -12$

Grading (2 P.): 1 P. formula $\bar{y} - \hat{\beta}_1\bar{x}$, 1 P. result -12 (with sign).

c) Interpretation of the slope

Interpret the slope $\hat{\beta}_1 = 3$ in one sentence in the context of the café.

$\hat{\beta}_1$ is the change of $\hat{y}$ when x increases by one unit. Here: one degree Celsius more goes along with, on average, 3 more iced coffees sold per day. The line describes an association within the

observed range (20–28 °C); it does not prove a causal effect, and it should not be extended far beyond the data.

Result: On a day that is 1 °C warmer, the café sells on average about 3 more iced coffees.

Grading (2 P.): 1 P. "per additional 1 °C", 1 P. "on average 3 more iced coffees".

Typical mistake: no units, or a causal claim ("the heat causes exactly 3 additional sales").

d) Python: prediction with `reg.predict`

The fitted regression of `sales` on `temp` is stored in the Python object `reg`; its estimated line is $\hat{y} = -12 + 3x$. Which value does the following line return?

```
reg.predict(pd.DataFrame({"temp": [25]}))
```

`reg.predict(...)` inserts the x-values of a new data frame into the estimated line. The new data frame needs a column with the *same name* as the explanatory variable in the formula, here `temp`:

```
reg.predict(pd.DataFrame({"temp": [25]}))      # 63.0
```

$$\hat{y}(25) = -12 + 3 \cdot 25 = -12 + 75 = 63.$$

$x = 25$ lies within the observed range $[20;\ 28]$, so this is an interpolation – no extrapolation problem.

Result: $\hat{y}(25) = -12 + 3 \cdot 25 = 63$: the command returns the predicted value 63 (iced coffees on a 25 °C day)

Grading (2 P.): 1 P. `predict` evaluates the estimated line at $x = 25$ ($-12 + 3 \cdot 25$), 1 P. result 63.

Typical mistake: reporting the coefficients (-12 and 3) instead of the prediction, or reading 25 as a number of iced coffees.

e) Residual of day 5

Compute the residual $\hat{u}_5$ of day 5 with $(x_5;\ y_5) = (28;\ 68)$, using the estimated regression line $\hat{y} = -12 + 3x$.

$$\hat{y}_5 = -12 + 3 \cdot 28 = 72, \qquad \hat{u}_5 = y_5 - \hat{y}_5 = 68 - 72 = -4.$$

On day 5 the café sold 4 iced coffees fewer than the line predicts; the point lies *below* the line. (All residuals: $-2,\ 0,\ +2,\ +4,\ -4$; they add up to 0.)

Result: $\hat{y}_5 = -12 + 3 \cdot 28 = 72,\ \hat{u}_5 = y_5 - \hat{y}_5 = 68 - 72 = -4$

Grading (2 P.): 1 P. fitted value $\hat{y}_5 = 72$, 1 P. residual -4 with the correct sign.

Typical mistake: residual with the wrong sign ($\hat{y}_5 - y_5 = +4$).

f) Coefficient of determination R^2

For the five days, the total sum of squares is $TSS = 400$ and the residual sum of squares is $RSS = 40$. Compute the coefficient of determination R^2.

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}} = 1 - \frac{40}{400} = 0.9 \qquad \left(= \tfrac{\text{ESS}}{\text{TSS}} = \tfrac{360}{400} \text{ with ESS} = \text{TSS} - \text{RSS} = 360\right).$$

90 % of the variation in the daily iced-coffee sales is explained (linearly) by the temperature; by the rule of thumb of the lecture ($R^2 \geq 0.8$) the fit is good. Check: $\text{ESS} = \hat{\beta}_1^2\, S_{xx} = 9 \cdot 40 = 360$ ✓

Result: $R^2 = 1 - \dfrac{\text{RSS}}{\text{TSS}} = 1 - \dfrac{40}{400} = 0.9$

Grading (2 P.): 1 P. formula $1 - \text{RSS}/\text{TSS}$ (or ESS/TSS with $\text{ESS} = 360$), 1 P. result 0.9.

Typical mistake: $R^2 = \text{RSS}/\text{TSS} = 0.1$ (share of the *unexplained* variation).

g) Correlation coefficient r_{xy}

Determine the correlation coefficient r_{xy} from $R^2 = 0.9$ and the slope $\hat{\beta}_1 = 3$.

In simple regression $R^2 = r_{xy}^2$, hence

$$|r_{xy}| = \sqrt{0.9} = 0.9487, \qquad r_{xy} = +0.9487$$

– positive, because the slope $\hat{\beta}_1 = 3$ is positive. Check with the sums: $r_{xy} = S_{xy}/\sqrt{S_{xx}S_{yy}} = 120/\sqrt{40 \cdot 400} = 120/126.49 = 0.9487$ ✓

Result: $r_{xy} = +\sqrt{0.9} = 0.9487$ (positive, because $\hat{\beta}_1 > 0$)

Grading (2 P.): 1 P. $|r_{xy}| = \sqrt{R^2}$, 1 P. value 0.9487 with positive sign.

Typical mistake: $r_{xy} = R^2 = 0.9$, or the sign ignored.

h) Python command (OLS)

The five days are stored in a pandas DataFrame `cafe` with the columns `temp` and `sales`. Give the `statsmodels` command that estimates the simple linear regression of `sales` on `temp`.

```
import statsmodels.formula.api as smf
reg = smf.ols("sales ~ temp", data=cafe).fit()
reg.params              # Intercept -12.0, temp 3.0
```

The dependent variable stands to the *left* of `~`, the explanatory variable to the right; `.fit()` carries out the least-squares estimation.

Result: `smf.ols("sales ~ temp", data=cafe).fit()`

Grading (2 P.): 1 P. `smf.ols(...).fit()`, 1 P. formula `"sales ~ temp"` (dependent variable on the left) with `data=cafe` (also accepted: the array form `sm.OLS(cafe["sales"], sm.add_constant(cafe["temp"])).fit()`).

Typical mistake: formula the wrong way round (`"temp ~ sales"`) or `.fit()` forgotten.

i) Ceteris-paribus interpretation

In the multiple regression of Part B, the coefficient of `temp` is $\hat{\beta}_1 = 3.0055$. Interpret this coefficient in one sentence.

In a multiple regression each coefficient is a *partial* effect: it compares days that differ in temperature but have the *same* rainfall.

```
Dep. Variable:                  sales   R-squared:                       0.947
Model:                            OLS   Adj. R-squared:                  0.945
Method:                 Least Squares   F-statistic:                     512.4
                  coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept    -12.2152      3.783     -3.229      0.002     -19.790      -4.641
temp           3.0055      0.131     22.863      0.000       2.742       3.269
rain          -0.4313      0.374     -1.154      0.253      -1.180       0.317
```

Model answer: "Holding the rainfall fixed, one degree Celsius more goes along with, on average, about 3.01 more iced coffees sold per day." The p-value 0.000 shows that this effect is highly significant.

Result: Holding the rainfall fixed, a 1 °C higher maximum temperature goes along with on average about 3.01 more iced coffees sold per day.

Grading (2 P.): 1 P. "+1 °C ⇒ on average about 3.01 (or 3) more iced coffees", 1 P. ceteris-paribus clause "holding the rainfall fixed".

Typical mistake: the clause "holding rain fixed" is missing – then the sentence describes the simple, not the multiple regression.

j) Significance of `rain`

In the multiple regression of Part B, the coefficient of `rain` *is* -0.4313 *with the p-value* 0.253 *(column* `P>|t|`*). Decide at* $\alpha = 0.05$ *whether rainfall has a significant effect on sales when the temperature is held fixed.*

t-test of a single coefficient: $H_0\colon \beta_2 = 0$ vs. $H_1\colon \beta_2 \neq 0$. Decision rule: reject H_0 if $p < \alpha$. Here

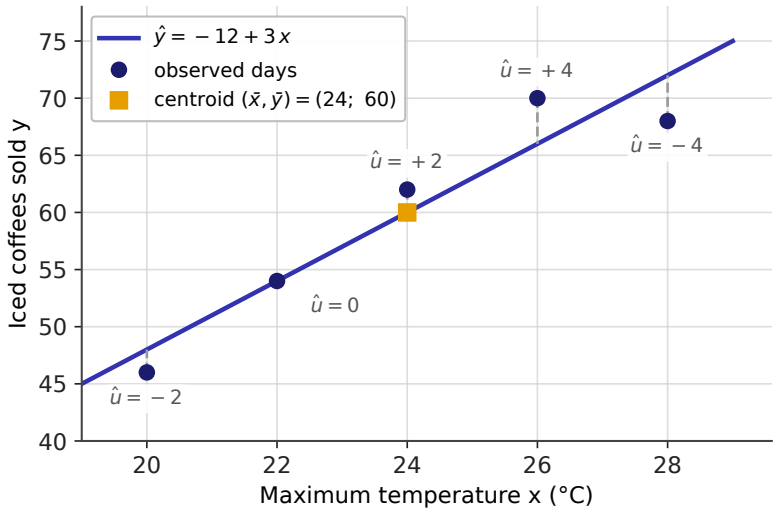
$$p = 0.253 \;>\; 0.05 = \alpha \quad \Rightarrow \quad \text{keep } H_0.$$

Equivalently, $|t| = 1.154 < t_{0.975;57} \approx 2.00$, and the 95 % CI $[-1.180;\ 0.317]$ contains 0. Once the temperature is in the model, the rainfall adds no significant explanation. This is *confounding*: alone, `sales ~ rain` gives a clearly negative slope (-6.19), because rainy days are also cooler days. "Not significant" does not prove "no effect": effects of about -1 iced coffee per mm are still compatible with the data.

Result: $p = 0.253 > 0.05 = \alpha \Rightarrow H_0\colon \beta_2 = 0$ is kept: given the temperature, rainfall has **no significant** effect on sales.

Grading (2 P.): 1 P. comparison $p = 0.253 > 0.05$, 1 P. conclusion "not significant / keep H_0".

Typical mistake: "$p > 0.05$, so rain has no effect at all" – not significant only means: no evidence found.



Scatter plot of the five days with the OLS line $\hat{y} = -12 + 3x$ and the residuals (dashed); the line passes through $(\bar{x}, \bar{y}) = (24;\ 60)$.

Typical mistakes in Task 2

- $\hat{\beta}_1 = S_{xx}/S_{xy}$ instead of S_{xy}/S_{xx}.
- Residual with the wrong sign ($\hat{y}_i - y_i$ instead of $y_i - \hat{y}_i$).
- $r_{xy} = R^2$, or the sign of r_{xy} ignored.
- Slope interpreted without units or as a proven causal effect.
- Multiple regression: the clause "holding the other variables fixed" forgotten; "not significant" read as "no effect".

4 Task 3 – Classification: logistic regression and evaluation (20 points)

Scenario. Churn of mobile-phone customers ($Y = 1$) explained by the number x of hotline calls: $\hat{\eta}(x) = -3 + 0.5\,x$. Confusion matrix of $n = 200$ test customers at $t = 0.5$: TN $= 140$, FP $= 20$, FN $= 10$, TP $= 30$.

↪ *Lecture reference:* Part 5, sections *Logistic Regression* (odds, log-odds, the model, odds ratio), *From Probabilities to Decisions* (classification rule) and *Evaluating a Classifier* (confusion matrix, threshold trade-off); Python: slide "Rates with Python: normalise the rows or the columns" and Summary 4.

The concepts behind this task

- **Odds:** $\text{odds} = \dfrac{p}{1-p}$; back: $p = \dfrac{\text{odds}}{1+\text{odds}}$.
- **Logistic model:** linear predictor $\eta = \beta_0 + \beta_1 x$ (log-odds), $p(x) = \dfrac{e^{\eta}}{1+e^{\eta}}$.
- **Odds ratio** e^{β_1}: factor by which the *odds* change per additional unit of x.
- **Classification rule:** $\hat{y} = 1$ if $\hat{p}(x) > t$; $t = 0.5 \Leftrightarrow \hat{\eta}(x) > 0$.
- **Confusion matrix:** accuracy $= (\text{TP}+\text{TN})/n$, sensitivity $= \text{TP}/(\text{TP}+\text{FN})$, specificity $= \text{TN}/(\text{TN}+\text{FP})$, precision $= \text{TP}/(\text{TP}+\text{FP})$.

a) Odds of churn

Last year, 20 % of all customers churned, i.e. $p = 0.2$. Compute the odds of churn.

$$\text{odds} = \frac{p}{1-p} = \frac{0.2}{0.8} = 0.25.$$

Odds of 0.25 mean "1 to 4": for every customer who churns, four customers stay.

Result: $\text{odds} = \dfrac{p}{1-p} = \dfrac{0.2}{0.8} = 0.25$

Grading (2 P.): 1 P. formula $p/(1-p)$, 1 P. result 0.25.

Typical mistake: odds confused with the probability (0.2) or computed as $(1-p)/p = 4$.

b) Churn probability at $x = 2$

Compute the predicted churn probability $\hat{p}(x)$ of a customer with $x = 2$ hotline calls, using $\hat{\eta}(x) = -3 + 0.5\,x$ and $e^{-2} = 0.1353$.

Pipeline $x \to \hat{\eta} \to \text{odds} \to \hat{p}$:

$$\hat{\eta}(2) = -3 + 0.5 \cdot 2 = -2, \qquad \text{odds} = e^{-2} = 0.1353, \qquad \hat{p}(2) = \frac{0.1353}{1+0.1353} = \frac{0.1353}{1.1353} = 0.1192.$$

A customer with two hotline calls churns with an estimated probability of about 12 %.

Result: $\hat{\eta}(2) = -3 + 0.5 \cdot 2 = -2$, $\hat{p}(2) = \dfrac{e^{-2}}{1+e^{-2}} = \dfrac{0.1353}{1.1353} = 0.1192$

Grading (2 P.): 1 P. linear predictor $\hat{\eta} = -2$, 1 P. probability 0.1192.

Typical mistake: reporting the log-odds -2 or the odds 0.1353 as the probability.

c) Interpretation of the odds ratio

The odds ratio of the hotline calls is $e^{\hat{\beta}_1} = e^{0.5} = 1.6487$. *Interpret it in one sentence.*

On the log-odds scale one call more adds $\hat{\beta}_1 = 0.5$; on the odds scale it *multiplies* the odds by $e^{0.5} = 1.6487$ – independent of the starting value of x:

$$\frac{\text{odds}(x+1)}{\text{odds}(x)} = e^{\hat{\beta}_1} = 1.6487.$$

Model answer: "Each additional hotline call multiplies the odds of churn by about 1.65 (the odds rise by about 65 %)." The change in the *probability* is not constant – it depends on where on the S-curve the customer is.

Result: Each additional hotline call multiplies the *odds* of churn by about 1.65, i.e. the odds of churn rise by about 65 % per call.

Grading (2 P.): 1 P. "per additional hotline call", 1 P. "*odds* multiplied by 1.65 / +65 %" ("probability +65 %": 0 P. for this part).

Typical mistake: "the probability of churn rises by 65 %" – the factor applies to the odds, not to the probability.

d) Classification at $x = 7$

A customer has $x = 7$ *hotline calls. Classify this customer with the threshold* $t = 0.5$, *using* $\hat{\eta}(x) = -3 + 0.5\,x$ *(you may use* $e^{0.5} = 1.6487$*).*

$$\hat{\eta}(7) = -3 + 0.5 \cdot 7 = 0.5, \qquad \hat{p}(7) = \frac{e^{0.5}}{1+e^{0.5}} = \frac{1.6487}{2.6487} = 0.6225.$$

Since $\hat{p}(7) = 0.6225 > 0.5 = t$, the customer is classified as **churner** ($\hat{y} = 1$). Shortcut: $t = 0.5$ corresponds to $\hat{\eta} > 0$, and $\hat{\eta}(7) = 0.5 > 0$. The decision boundary lies at $\hat{\eta} = 0 \Leftrightarrow x = 6$: customers with at least 7 calls are flagged.

Result: $\hat{\eta}(7) = 0.5 > 0$, $\hat{p}(7) = \dfrac{1.6487}{2.6487} = 0.6225 > 0.5 \Rightarrow \hat{y} = 1$: predicted to **churn**

Grading (2 P.): 1 P. $\hat{\eta} = 0.5$ or $\hat{p} = 0.6225$, 1 P. classification $\hat{y} = 1$ (churn).

e) Python command (logit)

The customer data are stored in a pandas DataFrame `df` *with the columns* `churn` *(coded* 0/1*) and* `calls`*. Give the* `statsmodels` *command that estimates the logistic regression of* `churn` *on* `calls`*.*

```
import numpy as np
import statsmodels.formula.api as smf
g = smf.logit("churn ~ calls", data=df).fit()
print(g.summary())      # z-tests of the coefficients
np.exp(g.params)        # odds ratios
```

`smf.logit` estimates the coefficients by maximum likelihood; the response must be coded 0/1.

Result: `smf.logit("churn ~ calls", data=df).fit()`

Grading (2 P.): 1 P. `smf.logit(...).fit()` (not `smf.ols`), 1 P. formula `"churn ~ calls"` with `data=df` (also accepted: the array form `sm.Logit(df["churn"], sm.add_constant(df["calls"])).fit()`).

Typical mistake: `smf.ols` instead of `smf.logit` (a linear probability model).

f) Accuracy

Compute the accuracy of the classifier from the confusion matrix with TN $= 140$, FP $= 20$, FN $= 10$ *and* TP $= 30$.

$$\text{accuracy} = \frac{\text{TP} + \text{TN}}{n} = \frac{30 + 140}{200} = \frac{170}{200} = 0.85.$$

85 % of all test customers are classified correctly (error rate 0.15). Note: the trivial rule "nobody churns" would already reach $160/200 = 0.80$ – accuracy alone is not enough.

Result: $\text{accuracy} = \dfrac{\text{TP} + \text{TN}}{n} = \dfrac{30 + 140}{200} = 0.85$

Grading (2 P.): 1 P. formula $(\text{TP} + \text{TN})/n$, 1 P. result 0.85.

g) Python: row shares with `pd.crosstab`

The DataFrame `test` *holds the actual class* `churn` *(coded* 0/1*) of the test customers,* `phat` *their predicted churn probabilities; at* $t = 0.5$ *the confusion matrix has* TN $= 140$, FP $= 20$, FN $= 10$ *and* TP $= 30$. *Which value does the following line show in the row* `churn` $= 1$, *column* `True`*?*

```
pd.crosstab(test.churn, phat > 0.5, normalize="index")
```

`normalize="index"` turns the counts into *row* shares: every row sums to 1, i.e. each entry is conditioned on the *actual* class. The row `churn` $= 1$ holds the 40 actual churners, the column `True` those flagged by $\hat{p} > 0.5$:

$$\frac{\text{TP}}{\text{TP} + \text{FN}} = \frac{30}{30 + 10} = 0.75 = P(\hat{y} = 1 \mid y = 1) = \text{sensitivity}.$$

```
col_0   False  True
churn
0       0.875  0.125
1       0.250  0.750
```

The rule catches 75 % of the customers who actually churn. The other row shows the specificity (0.875). `normalize="columns"` would condition on the prediction instead (column `True`: precision $30/50 = 0.6$).

Result: `normalize="index"` divides by the row total: $\dfrac{\text{TP}}{\text{TP} + \text{FN}} = \dfrac{30}{40} = 0.75$ – the sensitivity

Grading (2 P.): 1 P. row share (divide by the row total "Actual Yes" $= 40$), 1 P. value 0.75.

Typical mistake: dividing by the column total ($30/50 = 0.6$ – that is `normalize="columns"`, the precision) or by n ($30/200$).

h) Specificity

Compute the specificity of the classifier from the confusion matrix with TN $= 140$, FP $= 20$, FN $= 10$ *and* TP $= 30$.

Condition on the row "Actual No" (160 customers who stay):

$$\text{specificity} = P(\hat{y} = 0 \mid y = 0) = \frac{\text{TN}}{\text{TN} + \text{FP}} = \frac{140}{140 + 20} = 0.875.$$

87.5 % of the loyal customers are correctly left alone.

Result: $\text{specificity} = \dfrac{\text{TN}}{\text{TN} + \text{FP}} = \dfrac{140}{160} = 0.875$

Grading (2 P.): 1 P. formula $\text{TN}/(\text{TN} + \text{FP})$ (row "Actual No"), 1 P. result 0.875.

Typical mistake: $\text{TN}/(\text{TN} + \text{FN}) = 140/150$ (column instead of row).

i) Precision

Compute the precision of the classifier from the confusion matrix with TN $= 140$, FP $= 20$, FN $= 10$ *and* TP $= 30$.

Condition on the column "Pred. Yes" (50 flagged customers):

$$\text{precision} = P(y = 1 \mid \hat{y} = 1) = \frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{30}{30 + 20} = 0.6.$$

60 % of the customers flagged as churners really churn – 4 of 10 alarms are false alarms. Sensitivity and precision both have TP on top but divide by different totals.

Result: $\text{precision} = \dfrac{\text{TP}}{\text{TP} + \text{FP}} = \dfrac{30}{50} = 0.6$

Grading (2 P.): 1 P. formula $\text{TP}/(\text{TP} + \text{FP})$ (column "Pred. Yes"), 1 P. result 0.6.

Typical mistake: confusing precision with sensitivity $(30/40)$.

j) Lower threshold $t = 0.3$

To catch more churners, the provider lowers the threshold from $t = 0.5$ *to* $t = 0.3$. *Does the sensitivity increase or decrease?*

Lowering t flags every customer with $\hat{p}(x) > 0.3$ instead of $\hat{p}(x) > 0.5$: on the log-odds scale $\hat{\eta}(x) > \log(0.3/0.7) = -0.847$, i.e. $x \geq 5$ calls instead of $x \geq 7$. Every customer flagged before is still flagged, plus some more. Hence TP can only rise and FN only fall:

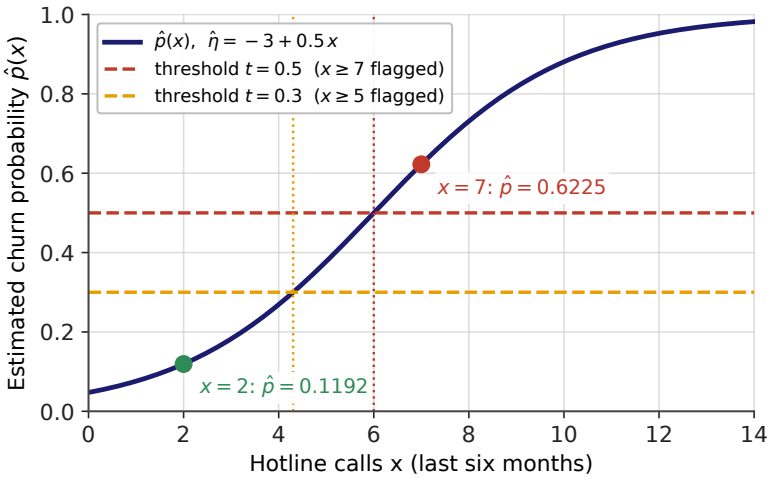
$$\text{sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \uparrow \quad \text{(denominator fixed at 40)}.$$

The price: more false alarms (FP ↑), so the specificity falls. Which threshold is best is a business decision (cost of a lost customer vs. cost of a retention offer).

Result: The sensitivity **increases** (it cannot decrease): with $t = 0.3$ more customers are classified as churners, so more of the actual churners are caught (TP ↑, FN ↓).

Grading (2 P.): 1 P. "increases", 1 P. reason: more customers predicted as churners ⇒ more actual churners caught.

Typical mistake: "decreases" – confusing sensitivity with specificity, which falls when the threshold is lowered.



Estimated churn probability $\hat{p}(x)$ with $\hat{\eta}(x) = -3 + 0.5\,x$: customers with $x = 2$ ($\hat{p} = 0.1192$) and $x = 7$ ($\hat{p} = 0.6225$); thresholds $t = 0.5$ (boundary $x = 6$) and $t = 0.3$.

Typical mistakes in Task 3

- Reading the log-odds $\hat{\eta}$ (or the odds $e^{\hat{\eta}}$) as a probability.
- "The odds ratio is 1.65, so the *probability* rises by 65 %."
- Mixing up sensitivity (divide by *actual* Yes, `normalize="index"`) and precision (divide by *predicted* Yes, `normalize="columns"`).
- Rows and columns of the confusion matrix confused (specificity $\neq \mathrm{TN}/(\mathrm{TN}+\mathrm{FN})$).
- Believing that a lower threshold catches *fewer* churners.